CHALMERS

CANDIDATE

DAT232-0009-KPD_2026-01-07_Bonus_NA

TEST

Ny tid - DAT232 VT26 LP2 Omtentamen 1 - J

Subject code	--
Evaluation type	--
Test opening time	10.01.2026 08:30
End time	10.01.2026 12:30
Grade deadline	--

PDF created	23.10.2026 11:38
Created by	Emil Skommargård

CHALMERS | GÖTEBORGS UNIVERSITET

DAT230/DAT232/DIT285 Advanced Requirements Engineering

Re-Exam

Wednesday, January 7th 2026

Examiner
Eric Knauss +46 31 772 10 80
Contact person during exam
Nayat Astaiza Soriano +46 70 464 89 43

Allowed tools / material
- English dictionary

General information
The exam consists of three parts:
- Part 1: 30 multiple choice questions, each with one correct answer and worth 1 Point.
- Part 2: One interactive question which is worth 10 Points
- Part 3: Four essay questions, each worth 10 Points. Maximal points can be given if:

  - The answer is correct.
  - The presentation of the answer is readable and clear.
  - The answer is given in English.

Grading
The grades on this exam are based on your total score on the questions:
0 – 39 points: Fail
40 – 55 points: 3
56 – 67 points: 4
68 – 80 points: 5

Review
Will be announced on Canvas.

Part 1: Multiple Choice Questions

This part consists of multiple-choice problems. These problems consist of pairs of Propositions and reasons. For each problem, you shall answer by using one of the following answers:

A Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
B Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
C The proposition is a true statement, but the reason is false.
D The proposition is false, but the reason is a true statement.
E Both the proposition and the reason are false.

Correctly answered problems give 1 point, while incorrect or missing answers give 0 points, regardless if you are partially correct in your answer!

1.1

Proposition	Reason
Requirements engineering is a systematic approach to reduce the likelihood to develop the wrong system, e.g. one that does not solve the problem of a user.	A requirement is a condition or capability needed by a user to solve a problem or achieve an objective.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.2

Proposition	Reason
It is crucial to satisfy only issues from important stakeholder groups when processing the raised issues from focus groups.	The stakeholder map should provide us with information about the power or influence that stakeholder groups have.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.3

Proposition	Reason
If a stakeholder appears to talk about too abstrac requirements, we should ask "how" questions.	"How" questions lower the level of abstraction and allow to discuss decomposing a high-level problem into solvable parts.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.4

Proposition	Reason
If a stakeholder uses unexpected or ambiguous wording during an interview, this should be followed up immediately and during the interview.	Missing information from an interview can easily be provided during analysis.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

1.5

Proposition	Reason
Questionnaires are a good elicitation technique in the early phases of a requirements engineering in a project.	Since a questionnaire does not allow for clarifying questions (as for example possible during interviews), it is important to carefully design each question in a questionnaire.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.6

Proposition	Reason
It's generally unnecessary to identify all events in the inner and outer domains.	Customers can generally validate a list of product events better than they can validate a list of domain events.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.7

Proposition	Reason
In practice, we have to live with tacit requirements.	In principle, we can assume that domain knowledge is non-obvious and specify all requirements in detail.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.8

Proposition	Reason
Tacit knowledge is easy to transfer to another person by writing it down.	When developers and users share a common language, it supports good communication.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.9

Proposition	Reason
According to Glinz, not all non-functional requirements are quality requirements.	Glinz distinguishes between Performance Requirements and Specific Quality Requirements, since the former are usually easy to operationalize, while the latter are usually hard to operationalize.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.10

Proposition	Reason
A development process requirement can be used when requirements are vague and time is short.	The sole purpose of development process requirements is to serve as a framework for structured requirements management.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.11

Proposition	Reason
Lauesen discourages using requirements as part of a contract between customer and supplier.	If requirements are part of a contract between customer and supplier, changing a requirement entails changing the contract, which may involve legal teams on both sides.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.12

Proposition	Reason
In order to achieve a good enough requirements specification, one should primarily focus on the requirements that everybody knows about.	Requirements that are well known by many stakeholders are easily misinterpreted.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.13

Proposition	Reason
Context diagrams can reflect an early decision about the scope of the product.	Context diagram arrows can refer to either data flow or events.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.14

Proposition	Reason
A data model is insensitive to the level of abstraction we work on.	If we make an accurate E/R model of the information occurring in the domain, the model will be similar with respect to these concepts to an E/R model of the data inside the final product (which may contain more detail).

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.15

Proposition	Reason
Feature requirements are useful when managing requirements for a large system.	Feature requirements are relatively short and independent descriptions of product functions and related data. Itis easy to store them in databases and they can have many administrative attributes.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propos ✔ n is correct.
- B) Both the proposition and the reason are correct statements, but the reason do not explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.16

Proposition	Reason
Customers may dream up so many features that the whole system becomes unrealistic. Feature requirements are hard to understand for the customer.	Customers can generally validate a list of product events better than they can validate a list of domain events.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propos n is correct. ✖
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.17

Proposition	Reason
"Enter guest name" is a good task description.	Good task descriptions are closed (they end with reaching a goal, with a pleasant feeling) and describe one session (a small set of related tasks).

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.18

Proposition	Reason
Task descriptions describe the interaction between a user and a system in detail.	The main scenario of a task descriptions usually alternates steps that a user executes with responses or actions that the system executes.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.19

Proposition	Reason
"Combinatorial creativity" combines both exploratory and transformational creativity.	A combinatorial creativity technique is generally a two-phase approach,where first, ideas are generated based on an exploratory technique, before these ideas are then discussed related to a transformed solution space.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. 

Correct. 1 of 1 marks.

1.20

Proposition	Reason
When relying on "Open Target" when specifying quality requirements, the customer indicates that the degree of fulfilment of this quality requirement is not important to them.	It is often difficult to decide the value needed for some quality attribute. "Open Target" gives the supplier the responsibility to investigate a suitable target value.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.21

Proposition	Reason
Within the quality grid, requirements engineers shall specify the concerns behind the priority of all the specified quality factors.	Quality grids are used to present the importance of quality factors.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.22

Proposition	Reason
The quality grid is useful to describe trade-offs between different quality requirements.	The quality grid provides a matrix of the relevant quality requirements. This allows to define trade-offs between two quality requirements in the cells.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.23

Proposition	Reason
Quality requirements are an exception to the rule that all requirements should be verifiable.	Since quality requirements are cross-cutting, criteria about whether they are met is better related only to the functional requirements they affect.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✖
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✔

Wrong. 0 of 1 marks.

1.24

Proposition	Reason
Requirements validation relates to the question "Are we building the right system".	Requirements validation is defined as the process of checking whether documented requirements match the stakeholders' needs.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✔
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.25

Proposition	Reason
Since it is important that all requirements are verifiable, it is implied that all requirements must be atomic.	If requirements were not clearly identifiable and would not allow to reason about them individually, it would be impossible to state whether they are met.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.26

Proposition	Reason
Consistency checks are done to compare different parts of a requirements specification, looking for inconsistencies.	CRUD checks are done to assess whether each entity in the data model is created, read, updated, and deleted,by an event, or a user task.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.27

Proposition	Reason
A prototype is only useful for verification, but cannot be used for validation.	A prototype only allows to answer whether we are building "the system correctly", but not whether we are"building the right system".

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.28

Proposition	Reason
Agile software development implies that development teams will have to do "Just-in-time" RE.	The only other option would be to fully analyse the requirements before the team starts to develop, effectively establishing a waterfall process.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✔
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.29

Proposition	Reason
Requirements engineering relates to the trade-off between common product goals and autonomy of agile teams.	The fine balance between decisions on portfolio, program, and team level must be maintained through strong communication of requirements-related knowledge.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✔
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✘
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.30

Proposition	Reason
User stories support requirements documentation as well as the agile principle of face to face communication.	When discussing user stories, it is important to add detailed results of that discussion to the documentation of the user story.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

Part 2: Interactive questions

2.1 This part of the exam consists of a match-assignment where different alternatives shall be matched with the most suitable, proposed, alternative. A correct answer gives 1 points, while incorrect or missing answers gives 0 points even if you are partially correct in your answer!

Match the ten problems described below with the most suitable alternative problem source. The same problem source may be the answer in more than one row of the table; some problem sources may not be the answer in any row. Only one problem source shall be selected on each row!

Please match the ten problems described below with the most suitable alternative problem source.

	Programm Error	Requirements Defect	Usability Problem
1) It is impossible to carry out the user registration task.	✗	✓	
2) The user erroneously believes that the registration task is completed.			✓
3) The user can register after lengthy attempts.			✓
4) The user registration functionality does not work as the programmer had intended.	✓		
5) The user cannot figure out which discount code to give the customer, although he knows which field to use.		✗	✓
6) The system does not provide the necessary fields to store user data.	✗	✓	
7) The system crashes when the user attempts to store user data.	✓		
8) The system does not offer an option to remove an item from the shopping cart.		✓	
9) The user is asked to include between 2 to 4 digits in the new password. The program only checks whether the password includes any digits.	✓		✗
10) The user thinks he has updated the password successfully, but the system requires the user to confirm the new password to complete the updating task.			✓

Partially Correct. 6 of 10 marks.

Part 3: Essay questions

For each topic below, write an essay that includes all of the listed concepts within the given maximum number of pages. The essays are graded based on

(a) how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and

(b) how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

3.1 For the topic below, write an essay that includes all of the listed concepts within the given maximum number of words. The essays are graded based on

1. how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and
2. how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

Please make an effort to write readable (e.g. by structuring your answer). Essays that are difficult to read or difficult to understand will render a deduction of received points. An unreadable essay will receive zero points! Each essay should start on a new sheet of paper.

Topic:
Difficulties of Elicitation **(10p)**

List of concepts:

- Gulfs of understanding
- symmetry of ignorance
- tacit knowledge
- stakeholder
- interviews

Maximum length: 750 words

Write your answer in the box below. Changes are saved automatically.

Difficulties of Elicitation:
Elicitation techniques in the requirement analysis is HARD process. It is very difficult to understand what stakeholder wants and their expectation and also what they are looking for. In some cases, Stakeholders may forgot to mention about the requirements in the requirement gathering stage, later it may cause the "Project failure". They are different types of elicitation techniques which is early phase, Mid phase, late phase. In each stage it has a complexity issues on their own way.

Gulfs of understanding:
It is a massive gaps analysis between stakeholder and developers. The main actors here is 'Stakeholder' and 'Developers'. Both has to understand their each domain and based on that elicitation should happen. If the project wants to be succeed, they should listen and properly align with the project goals. In many cases, Stakeholder may not align with the developers/supplier and later it may increases cost and time in the project.

It should be properly aligned in the each understanding level. Stakeholder clearly explain what they want,their expectations. Developer can start implementing the project based on the requirement finalized by the stakeholder.

For example., Stakeholder understanding about their business angle, Were Developers understanding about their technical side. Both actors has a different opinions in their own terms.

Symmetry of ignorance:
Both are the ignorant in their own way. Stakeholder doesn't understand the technical terms., likewise developers doesn't understand business terms, they will not understand their each other domain, at end it becomes the conflict between stakeholder and developer.Stakeholder should manage to express what they want and developer should able to manage it in process aligned to the project scope.It is very difficult to understand what stakeholder wants,

their understanding, what they are looking for. They often contradicted each other and it will rework the project.

Tacit knowledge:

In practice, We have to live with tacit requirements. They are knowledgeable person, but difficult to express it. we can assume that domain knowledge is non-obvious and specify all requirements in detail. Developers and users should share a common language, it supports good communication. Stakeholders may forgot to mention about the requirements in the requirement gathering stage, later it may cause the "Project failure". Mostly stakeholder didn't express what they wanting the beginning of the elicitation, like in the requirement gatherings stage. Over the period of time, they keep changeling the requirement and ended-up lot of ambiguity in the project development phase. In some case, Developers has to rework in the project, requirement keep changing all the time. Project launching date keeps drag, amount of cost is also increased, resources time will also increase. Multiple sprint meeting is required to complete the project scope.

Interviews:

Interviews are common in the elicitation techniques. It is very expensive and time consuming in the elicitation techniques. It can be structured interviews, Unstructured interviews,semi-structured interviews. In structured interviews, The questioned should be properly aligned to the needs, In Unstructured interviews is not well planned or not defined. In semi-structured interviews its is mix of both. If a stakeholder appears to talk about too abstract requirements, we should ask "how" questions. "How" questions lower the level of abstraction and allow to discuss decomposing a high-level problem into solvable parts.If a stakeholder uses unexpected or ambiguous wording during an interview, this should be followed up immediately and during the interview.Need to identified who is the right stakeholders for this project scope and it should be aligned in the proper way, In such case we can avoid rework in the project scope. It is crucial to satisfy only issues from important stakeholder groups when processing the raised issues from focus group. The stakeholder map should provide us with information about the power or influence that stakeholder groups have.

Words: 621

Answered.

3.2 For the topic below, write an essay that includes all of the listed concepts within the given maximum number of words. The essays are graded based on

1. how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and
2. how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

Please make an effort to write readable (e.g. by structuring your answer). Essays that are difficult to read or difficult to understand will render a deduction of received points. An unreadable essay will receive zero points! Each essay should start on a new sheet of paper.

Topic:
Data Requirements **(10p)**

List of concepts:

- Data model
- Data dictionary
- Virtual Windows
- Specify Quality of Data
- Relationship to Functional Requirements

Maximum length: 750 words

Write your answer in the box below. Changes are saved automatically.

Data model: (ER model)

It is conceptual model of describing the database. Data model can be defined and described as

"What is shows",

"what is stored",

"how it is structured".

A data model is insensitive to the level of abstraction we work on. If we make an accurate E/R model of the information occurring in the domain, the model will be similar with respect to these concepts to an E/R model of the data inside the final product (which may contain more detail).

It demonstrate ER model which Entity, Attributes, Relationships.

Entity - Object that define entity in the data model.

Attributes - Properties that describes the entity

Relationships - Connection of the defined entity.

Example: In the banking app, Customer own the account (entity) and that account holds the many transactions. ER model has a very high-level overview of the project scope to understand. It gives you the overview of the data,entity and their relationship to the project scope aligned.

Data dictionary:

In data model, It shows the overviews of the Data Requirements. In Data dictionary, it shows the detailed way of explaining the Data Requirements with the data format and with he clear description to understand the developers. The developer can easily understand the detailed of the requirements in this format so that they don't have any ambiguity in the project scope. This alignment really helps both stakeholder and developer are properly understand the requirement/scope.

Example: Customer says "I want unique 10 digit account numbers". So developer may thinks it may be a combination of both digits and character. Data dictionary should be defined in the proper format and descriptions.

Name: Account number

Type (ID): Integer

Description: Account number should be in 10 digits and it should be unique to all the users.

Virtual Windows:

Virtual windows defines the representation of the UI screens that user expected in the Data Requirements. It is mainly to clearly understand the user perspective to align the usability of the stakeholders in the environment.

Example: In the mobile banking systems, While demonstrating the wireframe of the system to the customer in the 'Transaction' screen, Customer says, "I wan to see the daily limit, so that i can track whether the payment failed or success and i also can track the transaction".

Specify Quality of Data:

Specify Quality of Data shows "how is the data is good enough" in the data requirement. It clearly define the "***Accuracy, Completeness, Timeliness, Consistency.***" These qualities brings the how the quality of data in the project scope this helps to defined the data in the specific manner. It is properly aligned to the requirements.

Example: In the mobile banking system, foreign exchange currency should be updated in every 60 seconds, without any disruptions.

Relationship to Functional Requirements:

All functional requirements should relationship to data model, data dictionary, Virtual windows and to the specify quality of data. Functional requirements are useful when managing requirements for a large system. Functional requirements are relatively short and independent descriptions of product functions and related data. It is easy to store them in databases and they can have many administrative attributes. It is linking to the goal traceability of the requirement. Quality requirements are an exception to the rule that all requirements should be verifiable.Since quality requirements are cross-cutting, criteria about whether they are met is better related only to the functional requirements they affect.

Words: 565

Answered.

3.3 For the topic below, write an essay that includes all of the listed concepts within the given maximum number of words. The essays are graded based on

1. how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and
2. how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

Please make an effort to write readable (e.g. by structuring your answer). Essays that are difficult to read or difficult to understand will render a deduction of received points. An unreadable essay will receive zero points! Each essay should start on a new sheet of paper.

Topic:

Managing Quality Requirements**(10p)**

List of concepts:

- Quality Grid
- Planguage
- Open Metric
- Relationship to Functional Requirements

Maximum length: 750 words

Write your answer in the box below. Changes are saved automatically.

Managing Quality Requirements:

Quality Grid:

Quality Grid in the quality requirements, is talk about " Prioritization". It mainly define what requirements to be prioritized. Quality grids are used to present the importance of quality factors. On which requirements customer is interested in, Their needs in way to make prioritize the requirements. Quality Grid helps in 'Usability' ' performance efficiency' in the quality requirements. Quality requirements are an exception to the rule that all requirements should be verifiable.

Example: In the banking systems,

"Security" is the critical.(If user doesn't find this is not secure, they will not use it).

"Speed" is also important. (Speed is important to the use, if it is slow, they don't feel like to use, But altogether, security of the system is more important than speed).

Planguage:

Planguage is the "Structure language" which has rich in information, clearly defined requirement.

Writing in Planguage is very important in managing quality requirements phase. It should be properly defined in the requirement and clearly understandable for the developers/tester to understand the requirement well. If it not properly well established in the requirement document, It ended-up in the confusion in the developing stage.

In many case, If the requirements was not properly aligned in the beginning, the project required the re-work in the later stage, it case many confusion, also it may lead to over cost of the project. so the developer keep changing it all the time. If requirementsis not aligned properly, In some case, Developers has to rework in the project, requirement keep changing all the time. Project launching date keeps drag, amount of cost is also increased, resources time will also increase. Multiple sprint meeting is required to complete the project scope

Example: Customer says., " I want system should be fast". What "fast"means in here. Planguage pushes to write in the proper way to define and make the developer to

understandable way.

Open Metric:

Open metric defines the "Performance metrics" and "Security metrics". These are two metrics is important in the Managing Quality Requirements. It is often difficult to decide the value needed for some quality attribute. "Open metric" gives the supplier the responsibility to investigate a suitable target value.

Example: In the banking system

In Performance metrics, The product should able to work on the certain stated conditions. It should work on the peak hours while multiple users uses the system. It should be 99% uptime so the users can able to use the product consistency.

In Security metrics, It should perform the authentication to use the product. well, it should not ask multiple authentication access to use the environment and the product should able to restrict the unauthorized access.

These Performance metrics and Security metrics should be properly aligned in the project scope that defines the managing quality requirements**.**

Relationship to Functional Requirements:

All functional requirements should relationship to entity, attributes and to the specify quality of data. Functional requirements are useful when managing requirements for a large system. functional requirements are relatively short and independent descriptions of product functions and related data. It is easy to store them in database and they can have many administrative attributes.

Words: 522

Answered.

3.4 For the topic below, write an essay that includes all of the listed concepts within the given maximum number of words. The essays are graded based on

1. how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and
2. how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

Please make an effort to write readable (e.g. by structuring your answer). Essays that are difficult to read or difficult to understand will render a deduction of received points. An unreadable essay will receive zero points! Each essay should start on a new sheet of paper.

Topic:
Checking and Validation **(10p)**

List of concepts:

- Contents check
- Structure check
- Consistency checks
- Reviews
- Tests

Maximum length: 750 words

Write your answer in the box below. Changes are saved automatically.

Checking and Validation:

Checking (Verification):

Requirements verification relates to the question ***"Are we building system right?!".***

Requirements verification is defined as the process of checking the product that fulfills the customers requirements. Requirements that are well known by many stakeholders are easily misinterpreted.Since it is important that all requirements are verifiable, it is implied that all requirements must be atomic. If requirements were not clearly identifiable and would not allow to reason about them individually, it would be impossible to state whether they are met.

There are few criteria that verification should follow:

Criteria: "Modified"

Description: "Has it been modified". It means the requirement can be modified.The Requirements will keep changing, it has the feasibility to modified at any stage.

Criteria: " Verified"

Description: "Has it been Verified". It means requirement can be verified.

Criteria: "Atomic"

Description: " Has to be specific to one idea". It means requirement has talk/ define about one specific idea.

Criteria: " Measured"

Description: "Has it been Measured"

Criteria: "Prioritize"

Description: "Has it been prioritized". It mainly define what requirements to be prioritized.

Validation:

Requirements validation is defined as the process of checking whether documented requirements match the stakeholders needs. Requirements validation relates to the question ***"Are we building the right system".***

The stakeholder map should provide us with information about the power or influence that stakeholder groups have. Questionnaires are a good elicitation technique in the early phases of a requirements engineering in a project. If requirements are part of a contract between customer and supplier, changing a requirement entails changing the contract, which may involve legal teams on both sides.

There are few criteria that validation should follow:

Stakeholders satisfactions: Does this requirements match the stakeholders needs?!

Risk assessment: Does this requirements has any risk in it?!

Goal assessment: Does this requirements liked to the goals aligned?!

Structure check &Consistency checks:

Structure checks are done to aligned in the proper way in the requirement. Writing in well defined requirement in the Planguage is very important in managing quality requirements phase. It should be properly defined in the requirement and clearly understandable for the developers/tester to understand the requirement well. If it not properly well established in the requirement document, It ended-up in the confusion in the developing stage.

In many case, If the requirements was not properly aligned in the beginning, the project required the re-work in the later stage, it case many confusion, also it may lead to over cost of the project. so the developer keep changing it all the time.

Consistency checks are done to compare different parts of a requirements specification, looking for inconsistencies. ***CRUD checks*** are done to assess whether each entity in the data model is created, read, updated, and deleted,by an event, or a user task.

Words: 460

Answered.